

Sensorless External Force Detection Method for Humanoid Robot Arm based on BP Neural Network

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Abstract—In order to improve the safety performance of the robot, this paper proposes an external force detection method for humanoid robot arm without using joint torque sensors, which can detect the external force of the joint space in real time during the operation of the robot. First, Analyses on the structure of the humanoid robot arm is performed, and the model of robot external force detection is established based on robot dynamics and motor dynamics. Then, the error of detection model is analyzed, and the robot dynamic model error is compensated by using the BP network, in order to obtain more accurate external force detection value of the robot. Experiments show that the method can effectively improve the detection accuracy, and the obtained external force detection data can be applied not only to the collision detection in the static state of the robot, but also to the online collision detection when the robot is running, which can ensure the safe operation of the robot.

I. INTRODUCTION

In recent years, thanks to the rapid development of technology, the application field of robots has extended from industrial production to the rescue[1], resource exploration [2], medical care[3, 4] and home service[5] fields. The environment has also changed from the regular industrial pipeline to complex and changeable real environments. For example, when intelligent service robots need to provide services for the crowd, and the human-robot cooperation and human-computer interaction become more frequent, which puts forward a higher requirement for the safe operation of the robot. In order to ensure the safety of the robot and improve the maneuverability of the robot, the robot should have the ability to quickly and accurately detect the external forces and give an appropriate response to them[6].

The external force detection of robots is the main direction of robot safety performance research. A lot of work on the external force detection method of robots has been conducted and many good research results have been achieved. Generally, according to whether the torque sensor is used or not, the external force detection methods can be divided into two categories: detection with sensor and sensorless detection.

External force detection methods with sensor[7, 8] can guarantee the real-time and accuracy of external force detection, they greatly increase the complexity of the system and the high cost of these sensors also limits their applications. Therefore, much work has also focused on the sensorless moment detection method.

Yao et al.[9] can obtain the approximate position and direction of determining the occurrence of collision based on the decoupling performance of the generalized momentum of the manipulator and the external torque. However this method cannot detect the value of the contact force, so it can only be used for collision detection, but not be extended to compliance control based on external force. Monfaredi et al.[10] proposed a method for estimating the external force of a surgical robot based on dynamics and motor current, and realized the torque estimation of the joint through dynamic parameter identification. To realize the detection of external force, Yamada et al.[11] tested the change of joint moment by detecting the current of driving motor and the speed of the robot without moment sensor, and compares it with the moment obtained by inverse dynamic model without collision. Villagrossi et al.[12] used the robot's thermal friction model combining with the task-oriented dynamics model to construct a virtual force observer, which was successfully applied in the robot polishing task. At present, a large number of researchers are studying the external force detection method of torque-free sensor robot based on system momentum[13-16]. However, the dynamic model of the system is needed here, and the detection accuracy is closely related to the accuracy of modeling. Unfortunately, the robot is a complex non-linear system, and there are many non-linear factors that cannot be accurately modeled, such as joint flexibility, transmission clearance, and nonlinear friction. How to compensate the nonlinear friction of the system and improve the accuracy of external force detection is the main work of this paper.

II. DESCRIPTION OF HUMANOID ROBOT ARM

A. Structure design of robot arm

The research in this paper is carried out on the self-developed intelligent humanoid robot system in the laboratory. Fig. 1 shows the intelligent humanoid robot we developed and its arm structure. The robot is developed to be used as comprehensive robot research platform on which the research activities of various robotic fields can be conducted. It is a half-body humanoid robot mounting on an omnidirectional moving platform, which consists of a 2-DOF waist, a 2-DOF head, two 7-DOF robot arms of human-like structure, and two 5-finger bionic hands. Our work is performed on the humanoid robot arm and the rest of robot are not our concerns. According to the configuration of the humanoid manipulator, the center of the shoulder of the robot is taken as the origin, and the Denavit-Hartenberg (DH) coordinate method is used to establish the linkage

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coordinate system of the robot arm. Table 1 shows the DH coordinate parameters of the robot, according to which the rotation matrix of the robot and the Jacobian matrix can be calculated.

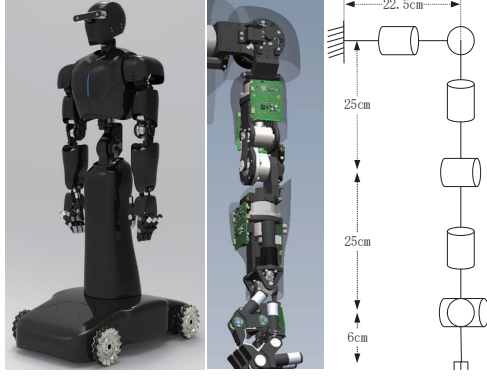


Figure 1. The humanoid robot and its arm structure

TABLE 1 THE DH PARAMETERS OF THE HUMANOID ROBOT ARM

Link i	a_{i-1}/m	$\alpha_{i-1}/^\circ$	d_i/m	$\theta_i/^\circ$
1	0	0	0.225	-90
2	0	-90	0	-90
3	0	90	0.25	90
4	0	90	0	0
5	0	-90	0.25	90
6	0	90	0	90
7	0	90	0	-90

TABLE 2 THE JOINT PARAMETERS OF THE HUMANOID ROBOT ARM

Joint	P_e (W)	C_i (mNm/A)	B_m (mNm/rmp)	I (A)	N_0 (rmp)	i	η
1	100	53.4	0.00561	4.43	3840	80	0.95
2	100	53.4	0.00561	4.43	3840	80	0.95
3	70	46.1	0.0021	2.73	6450	80	0.95
4	70	46.1	0.0021	2.73	6450	80	0.95
5	50	46.1	0.0021	1.61	6080	80	0.95
6	20	30.8	0.0021	1.5	9690	159	0.93
7	20	30.8	0.0021	1.5	9690	159	0.93

Note: i means the joint reduction ratio; B_m means the damping coefficient of the motor; η means the transmission efficiency of the harmonic reducer.

B. Structural Design of Robot Joints

For the robot without torque sensors, the joint structure design is crucial for the realization of robot external force detection. In the absence of dedicated sensors, the real-time motion states of the robot must be obtained only by the current motion states of the motor and the operating states of the motor, and the joint design of the robot needs to meet the requirements of small reduction ratio and light load. Meanwhile a high-precision speed feedback and an angle feedback should be set at the joint of the robot to provide data for sensorless torque detection.

In the design of the manipulator, for each of the first five joints, a DC brushless motor and a harmonic reducer are combined to form a rotating joint. The other two joints at the end of the manipulator are respectively constructed by connecting two DC geared motors in parallel. Table 2 shows

the motor parameters and reduction ratios for these seven joints.

III. MATHEMATICAL MODEL OF EXTERNAL FORCE DETECTION

For lack of a torque sensor, the current actual joint drive torque can be calculated from the motor current data and compared with the theoretical joint torque value of the robot under the current motion state to obtain the external force of the joint space, and the external force detection model of the robot can be expressed as:

$$\tau_{ext} = \tau_{em} - \tau_{md} \quad (1)$$

where τ_{ext} is the external force, τ_{em} is the joint drive torque, and τ_{md} is the theoretical joint torque.

A. Joint Driving Force Model

In the robot system, the driving torque of the joint is essentially the effect the load torque on the connecting rod which is provided by the joint motor and the reduction gear. The motion equation of the motor can be expressed as follows:

$$T_e - T_L - B_m \omega = J \frac{d\omega}{dt} \quad (2)$$

where ω is the rotational speed of the joint drive motor, T_L is the load torque of the joint motor, and the joint speed q can be obtained by dividing the joint reduction ratio, B_m is the damping coefficient of the motor, and J is the rotor inertia moment of the motor. The electromagnetic torque of the brushless DC motor T_e is generated by the interaction between the stator current and the rotor magnetic field:

$$T_e = (e_a i_a + e_b i_b + e_c i_c) \frac{1}{\omega} \quad (3)$$

At any time during the operation of the motor, only when two phases of the stator are turned on and the induction electromotive force and the stator current are strictly synchronized in phase, the constant electromagnetic torque can be generated. The motor electromagnetic power can be expressed as:

$$P_e \approx e_a i_a + e_b i_b + e_c i_c = 2e_s i_s \quad (4)$$

where e_s is the induced electromotive force, i_s is the stator current's RMS, so there is:

$$T_e = \frac{P_e}{\omega} = \frac{2e_s i_s}{\omega} \quad (5)$$

The induction electromotive force of the motor is determined by the motor speed ω , the magnetic flux density B_f and the motor induced electromotive force coefficient C_e :

$$e_s = C_e B_f \omega \quad (6)$$

so there are:

$$T_e = 2C_e B_f i_s \quad (7)$$

For the permanent magnet brushless DC motor used in this paper, the magnetic flux is generated by the permanent magnet as a constant value, so the electromagnetic torque of the brushless DC motor can be rewritten as:

$$T_e = C_{tm} i_s \quad (8)$$

where C_{tm} is the motor torque coefficient which is generally indicated in the motor instruction manual. The above Table 2 lists the values of the motors used the robot.

The load torque of the motor passes through the harmonic reducer and then drive the link. By substituting the torque transfer parameters of the harmonic reducer, the driving torque of each joint is:

$$\tau_{em} = i\eta T_L \quad (9)$$

where i is the reduction ratio, and η is the transmission efficiency which is related to the operating state of the reducer.

From (2), (8) and (9), we can get the actual output torque of the joint:

$$\tau_{em} = i\eta C_{tm} i_s - i\eta B_m \omega - i\eta J \dot{\omega} \quad (10)$$

B. Dynamics Model

Since the robot arm the studied in this paper has seven degrees of freedom, when the dynamic model of the robot arm is established, it will involve complex kinematic coupling relations, which will greatly increase the solving difficulty. Due to the same solving principle of each joint in the process of dynamic solution of the system, only the first and fourth joints of the robot are used and the rest joints are locked so as not to rotate, so that the robot arm can be transformed into a simpler two-link model which can be used to illustrate the proposed method in this paper. Fig. 2 is the simplified model.

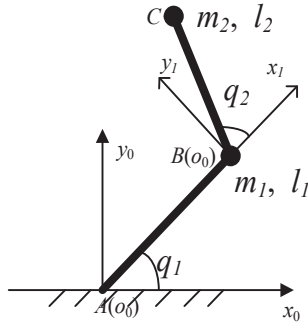


Figure 2. The simplified 2-link model

The Lagrangian method is used to model and obtain the theoretical value of the dynamic model of the joint drive torque:

$$\tau_{md} = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) \quad (11)$$

$$M(q) = \begin{bmatrix} (m_1 + m_2)l_1^2 + m_2l_2^2 + 2m_2l_1l_2 \cos q_2 & m_2(l_2^2 + l_1l_2 \cos q_2) \\ m_2l_1l_2 \cos q_2 & m_2l_2^2 \end{bmatrix}$$

$$C(q, \dot{q}) = \begin{bmatrix} -2m_2l_1l_2 \sin q_2 & -m_2l_1l_2 (\dot{q}_1 + \dot{q}_2) \sin q_2 \\ m_2l_1l_2 \dot{q}_1 \sin q_2 & 0 \end{bmatrix}$$

$$G(q) = \begin{bmatrix} (m_1 + m_2)gl_1 \sin q_1 + m_2gl_2 \sin(q_1 + q_2) \\ m_2gl_2 \sin(q_1 + q_2) \end{bmatrix}$$

where q is the rotation angle vector of $n \times 1$, M is the positive definite inertia matrix of $n \times n$, C is the centrifugal force and Coriolis force matrix of $n \times n$, G is the gravity matrix of $n \times 1$, and n is the degree of freedom of the joint, here n is 2.

IV. MODEL ERROR COMPENSATION

It can be seen from above that the external force acting on the machine can be obtained by solving the joint driving torque and dynamic model. However, the above dynamic model of the robot does not take into consideration of the effects of various nonlinear factors that actually exist, such as friction, joint recoil and joint clearance. So the external force detection method should be revised as:

$$\tau_{ed} = \tau_{em} - \tau_{md} - \tau_{cm} \quad (12)$$

where τ_{cm} is the compensation value of the robot kinetic model. This paper will compensate the modeling error of (11) through BP network.

A. Mechanism Model Error

The theoretical driving torque and the actual joint torque in (1) can be obtained by the previous modelling process. The error of robot joint model can be obtained by experiment as follows. Fig. 3 shows the output of the actual driving torque and theoretical torque of the joint when joint 1 is -90° and joint 2 changes from 0° to 90° without external disturbance.

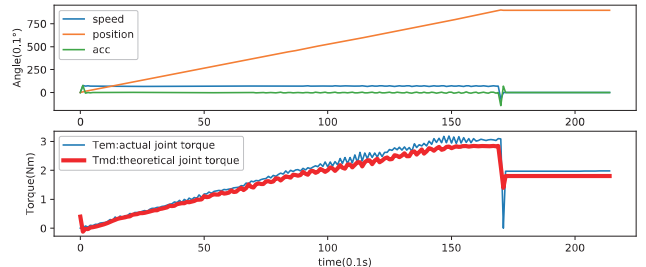


Figure 3. The error of robot joint model

In Fig. 3, *speed* is the rotation speed curve of joint 1, *acc* is the change curve of the joint acceleration, *position* is the change curve of joint angle, T_{em} is the actual output torque of the joint, and T_{md} is the theoretical torque of the joint obtained through the robot dynamics model. According to the model theoretically, in the absence of external force interference, the result that the external force is zero should be obtained, that is, the T_{em} and T_{md} curves should coincide with each other. However, from the experimental results, it can be seen that there is a significant error between the T_{em} and T_{md} curves, especially at the start and stop moments.

This is due to the fact that various nonlinear factors are not considered in the modeling of T_{md} .

Due to lack of a nonlinear model of the mechanism, we uses a data-driven supervised learning method to model nonlinear terms, and a multi-layer perceptron neural network is trained to predict the model error. To train the model to get better results, first of all, a large number of training data need to be obtained. In order to ensure the generalization ability of the model, all the data should be evenly distributed in the whole motion space. In our experiments, the motion range of joint 1 is limited to $0-108^\circ$, and the motion range of

joint 2 is 0-120°. We obtained the data of 13 points uniformly distributed in the motion range of each joint by keeping one joint still while the other joint moves forward or backward in the entire range of motion. Each obtained input sample data during the motion includes the angle, angular velocity and angular acceleration values and the kinetic model errors are calculated as the sample output.

A total of 56 sets of sample data were obtained, in which 40 were used as training sets to train the network, and 16 were taken as test sets to test the trained models and evaluate the training effects.

B. Establishment of BP Neural Network Structure

The BP algorithm is used to train the multi-layer perceptron network and realize the compensation of mechanism error. The network is set to have one input layer,

one output layer and two hidden layers at first. The input layer has 6 neurons including the angle, angular velocity and angular acceleration values of the two motor joints. The output layer has 2 neurons which are respectively the model errors of theoretical torque of the two joints. Since the fitting ability of different network structures varies greatly, so an experiment on the network structure adjustment is conducted to verify which network structure is suitable for the application of this method. By continuously changing the number of hidden layers and the number of neurons in each hidden layer, the network model output loss of different structures is calculated on the training set and the test set so as to compare the fitting abilities of different network structures. The loss function is defined as the mean square error (MSE). Table 3 shows some experimental results on the network structure.

TABLE 3 TRAIN RESULTS OF DIFFERENT NEURAL NETWORK

Number of hidden neurons				Loss				Number of hidden neurons				loss				Number of hidden neurons				Loss			
hidden1	hidden2	train	Test	hidden1	hidden2	train	test	hidden1	hidden2	train	Test	hidden1	hidden2	train	test	hidden1	hidden2	train	Test	hidden1	hidden2	train	Test
6	0	0.034	0.0347	100	4	0.0319	0.0317	400	0	0.0017	0.0041	100	0	0.039	0.0363	500	0	0.0013	0.0013	100	0	0.0086	0.0103
10	0	0.039	0.0363	100	20	0.0246	0.0245	100	50	0.0261	0.0254	80	0	0.0079	0.0115	100	50	0.0337	0.0336	80	80	0.0039	0.0043
50	0	0.0086	0.0103	100	5	0.026	0.0256	50	80	0.0066	0.0077	150	0	0.0068	0.0086	80	80	0.0042	0.0046	200	0	0.0059	0.0077
80	0	0.0079	0.0115	20	5	0.0322	0.03232	80	0	0.0041	0.0063	250	0	0.0055	0.0088	50	0	0.0041	0.0063	20	20	0.0256	0.0256
100	0	0.0079	0.108	20	20	0.0256	0.0256	80	0	0.063	0.0073	20	80	0.023	0.224	80	0	0.063	0.0073	20	80	0.023	0.224
150	0	0.0068	0.0086																				
200	0	0.0059	0.0077																				
250	0	0.0055	0.0088																				

As seen from the table, the best fitting effect can be obtained when the network has one single hidden layer and the number of neurons in the hidden layer is 500. The error compensation network of this structure with its parameters are used for the online prediction of robot external force. And thus a BP neural network for the model error compensation has been established. This BP network needs to input the current running state of the robot, and the output is the nonlinear error compensation term of the model, τ_{nn} .

After the compensation by the neural network method, the final external force detection model can be expressed as:

$$\tau_{ext} = \tau_{em} - \tau_{md} - \tau_{nn} \quad (13)$$

V. EXPERIMENTS AND ANALYSES

The two experiments are performed to test the effectiveness of the proposed external force detection method. One is an external force detection accuracy test to verify the accuracy of the robot external force detection method; the other is a robot online collision detection test to verify the real-time performance of the method. Experiments were performed on the humanoid robot platform described in Section 2.

A. External Force Detection Accuracy Test

In this test, a high-precision dynamometer for detected external force reference is used to verify the accuracy of the proposed method. The reference force is acquired by holding the dynamometer to push the robot in a manual way. By comparing the external force value of the robot and the

output of the dynamometer, the detection accuracy of the method can be verified. In particular, the external force detection output value of the robot system is torque, and the output of the dynamometer is an external force value, so the torque value needs to be divided by the length of the link before the comparison can be made.

In order to improve the test accuracy, the forces of joint 1 and joint 2 need to be decoupled, that is, the upper arm and forearm of the robot should be kept in a vertical state. Then, the accuracy of external force detection can be verified, respectively by carrying out four groups of tests as shown in Fig. 4.

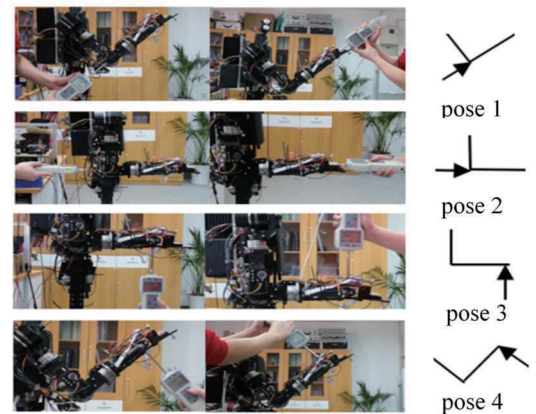


Figure 4. External force detection accuracy test

In order to keep the consistency in the output data of two detection ways, the detected peak force values of the dynamometer output and the system output (our method) are used as the test results. Each pose was sampled 5 times in both two directions.

The experimental results are shown in Table 4, in which our neural network method are compared with the compensation method based on Strebeck friction model[16]. In the table, M is the peak value of the output of the

dynamometer, T_{ni} is the peak value of the output by the neural network compensation method, and E_n is the corresponding detection error; T_{sl} is the output for the output with the Strebeck friction model, and E_s is the corresponding detection error. It is noted that the external force detection output value of the robot system is torque, and the output of the dynamometer is force, so the torque value needs to be change into the force value by being divided by the length of the link before the comparison can be made and the unit of each variable is N.

TABLE 4 RESULTS OF EXTERNAL FORCE DETECTION ACCURACY TEST

Pose1					Pose2				
T_{sl}	T_{n1}	M	E_s	E_n	T_{sl}	T_{n1}	M	E_s	E_n
23.19	22.19	21.94	1.25	0.25	41.75	42.76	42.74	-0.99	0.02
33.29	34.35	34.14	-0.85	0.21	40.05	39.66	39.81	0.24	-0.15
23.34	22.38	22.7	0.64	-0.32	34.21	34.48	34.56	-0.35	-0.08
39.37	38.74	38.5	0.87	0.24	29.23	30.11	29.87	-0.64	0.24
34.22	34.14	34.56	-0.34	-0.42	23.81	23.06	23.24	0.57	-0.18
-19.48	-17.91	-18.25	-1.23	0.34	-36.55	-37.44	-37.8	1.25	0.36
-26	-25.26	-25.21	-0.79	-0.05	-37.57	-38.51	-38.24	0.67	-0.27
-9.17	-10.06	-10.21	1.04	0.15	-35.68	-36.89	-37.28	1.6	0.39
-22.57	-24	-23.65	1.08	-0.35	-27.74	-28.06	-28.14	0.4	0.08
-36.66	-36.56	-36.42	-0.24	-0.14	-27.02	-27.41	-27.56	0.54	0.15
Pose3					Pose4				
T_{s2}	T_{n2}	M	E_s	E_n	T_{s2}	T_{n2}	M	E_s	E_n
37.98	37.15	36.94	1.04	0.21	26.66	26.24	26.12	0.54	0.12
30.23	30.34	30.48	-0.25	-0.14	27.85	28.45	28.2	-0.35	0.25
27.38	26.2	26.54	0.84	-0.34	31.82	30.25	30.54	1.28	-0.29
30.35	30.8	30.81	-0.46	-0.01	38.64	39.82	39.58	-0.94	0.24
26.58	27.48	27.23	-0.65	0.25	29.73	28.84	29.21	0.52	-0.37
-21.76	-22.32	-22.6	0.84	0.28	-25.25	-25.74	-25.94	0.69	0.2
-34.84	-35.68	-35.82	0.98	0.14	-37.46	-36.97	-37.12	-0.34	0.15
-43.6	-44.35	-44.27	0.67	-0.08	-37.76	-37.01	-36.82	-0.94	-0.19
-50.08	-49.33	-49.54	-0.54	0.21	-34.89	-35.88	-35.54	0.65	-0.34
-25.19	-25.05	-25.43	0.24	0.38	-17.35	-17.25	-17.59	0.24	0.34

It can be seen that the average error of the neural network method is smaller than that of the friction model. Therefore, compared with the method based on mechanism modeling and parameter identification by friction, the neural network method has better model error compensation ability, which verifies the effectiveness of this method. It can also be seen from the above table that the errors of joint 1 are generally larger than that of joint 2, but the maximum error is not more than 0.1Nm, which can be considered as the error caused by the current noise of motor.

B. Collision Detection Experiment

In order to check the effectiveness and real-time of the external force detection of the robot under motion, a wooden stick was placed in the path of the robot arm, and we experimented with opening the collision response and not opening the collision response. The collision response strategy adopts the reflection control strategy, it means the external force is detected and then retracted in the direction of the external force. The collision experiment process without collision response is shown in Fig. 5. The angle P_1 of joint 1 and the external force detection value T_1 during the experiment were recorded, and the experimental data is

shown in Fig. 6. From the experimental results, it can be seen that if there is no collision detection and an obstacle occurs in the robot running path, a violent collision will occur, which will cause damage to the robot or the object as shown in Fig. 5.

In contrast to this, a collision test with a collision response is performed, and the external force detection value of joint 1 is used as a trigger switch. When $T_1 > 0.5\text{Nm}$, the collision is considered to occur and the arm is moved away from the collision position. The experimental process is shown in Fig. 7. The angle P_1 of joint 1 and the external force detection value T_1 during the experiment were recorded, and the experimental data is shown in Fig. 8.

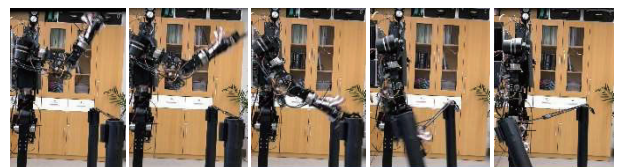


Figure 5. Collision experiment process without collision response

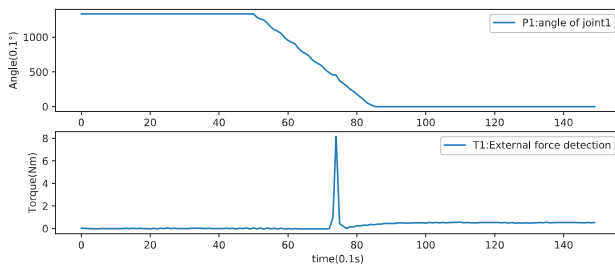


Figure 6. Collision experiment data without collision response

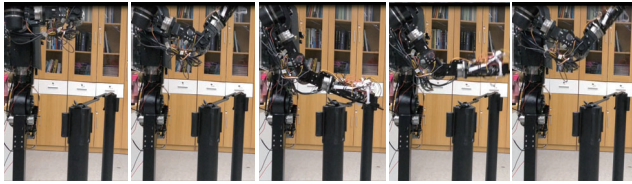


Figure 7. Collision experiment process with collision response

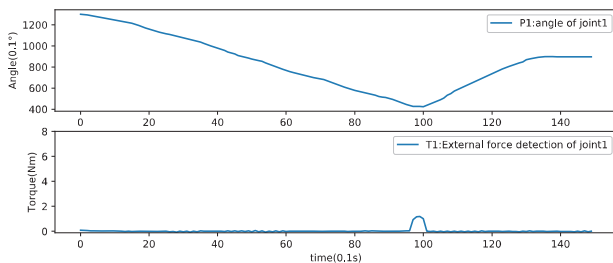


Figure 8. Collision experiment data with collision response

It can be seen that after the collision response is turned on, the robot can detect the occurrence of the collision in time and leave the collision position to avoid the occurrence of injury, and the force at the time of the collision is greatly reduced. It can be seen that the wooden rod is broken when the collision response is not turned on, and the maximum external force value reaches 8Nm, which is likely to cause serious damage to the robot body and the environment; After the collision response is turned on, the maximum external force value is only 1.2Nm, and the wooden stick is not broken. This shows the effectiveness and real-time performance of the proposed method.

VI. CONCLUSION

In this paper, we use the excellent nonlinear function fitting ability of neural network to compensate the nonlinear part of the robot mechanical model which is difficult to accurately model. Experiments show that this method can effectively improve the compensation accuracy compared with the traditional friction mechanism modeling compensation method. The method can ensure real-time performance and can perform online external force detection during the operation of the robot. The occurrence of collision is detected, which provides data support for the robot's safe

collision response strategy, thus ensuring the safety of the robot during operation.

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